



Task 2: Neural Networks

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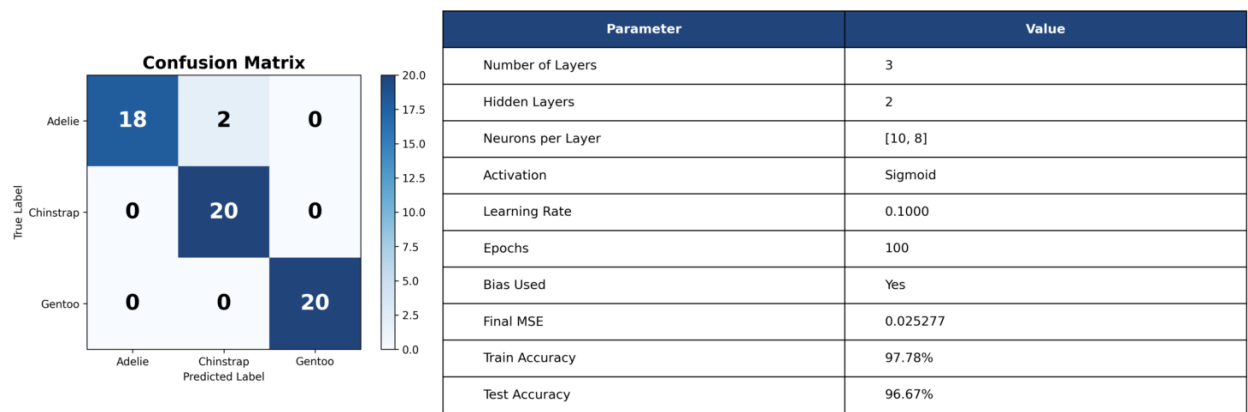
Experiments:

We conducted twelve experiments varying the number of epochs, learning rates, the use of bias, and the choice between tanh and sigmoid activation functions.

- Trying 2 hidden Layers, Epochs = 100, Learning Rate = 0.1 and with Bias.

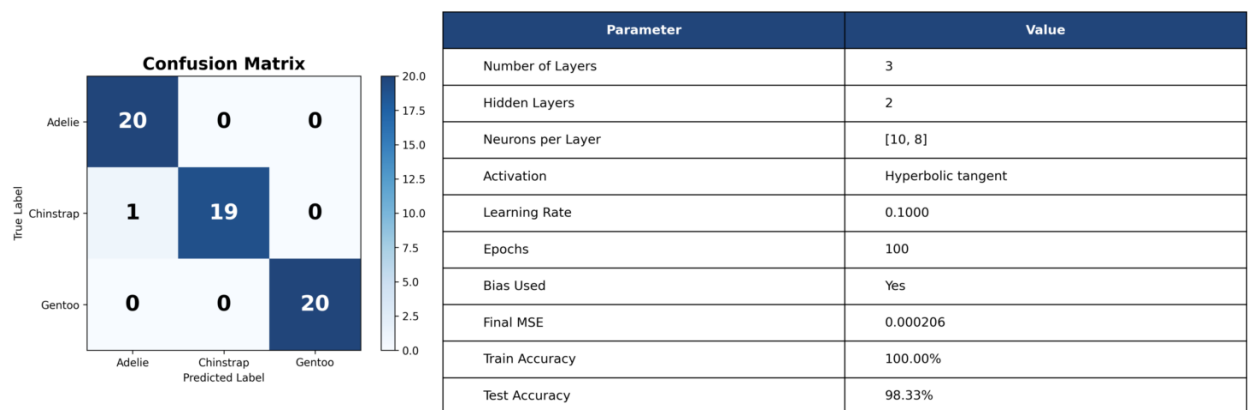
Sigmoid:

Penguin Species Classification - Backpropagation Results



Tanh:

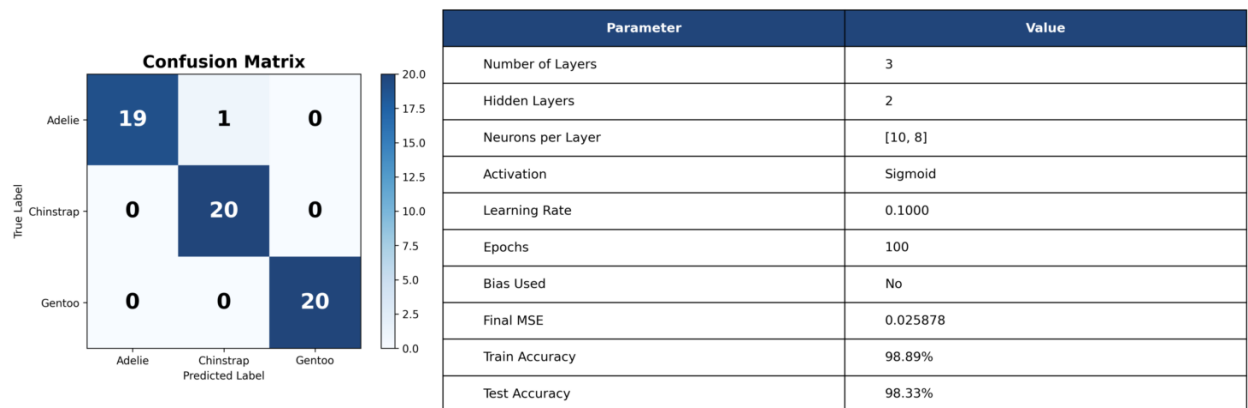
Penguin Species Classification - Backpropagation Results



- Trying 2 hidden Layers, Epochs = 100, Learning Rate = 0.1 and without Bias.

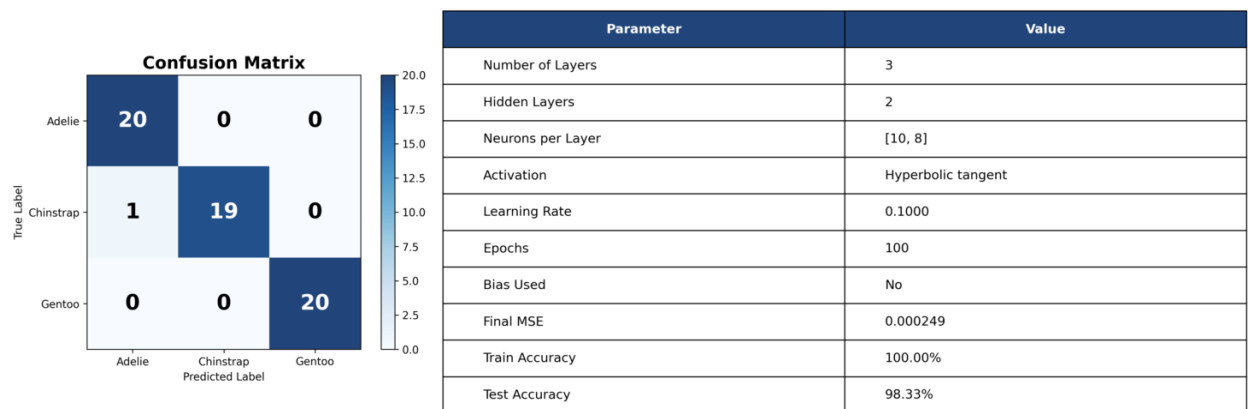
Sigmoid:

Penguin Species Classification - Backpropagation Results



Tanh:

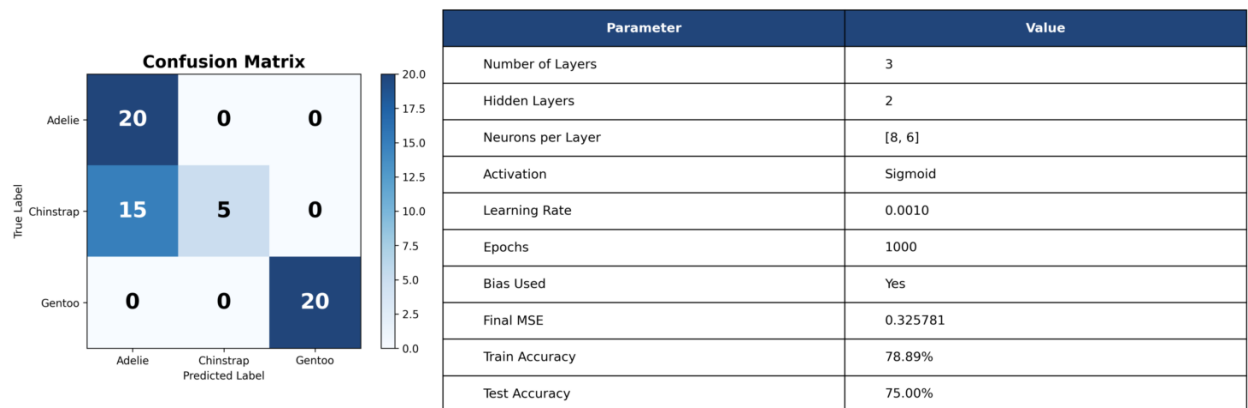
Penguin Species Classification - Backpropagation Results



- Trying 2 hidden Layers, Epochs = 1000, Learning Rate = 0.001 and with Bias.

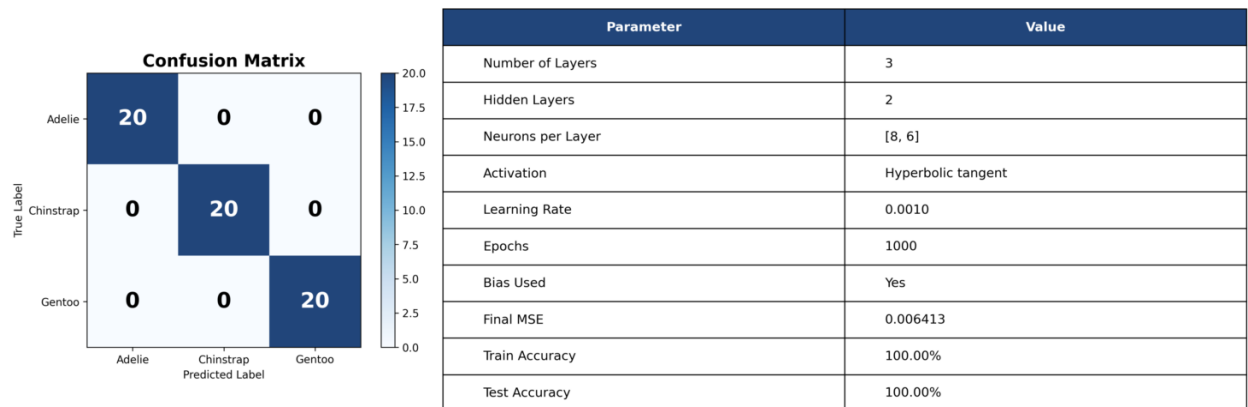
Sigmoid:

Penguin Species Classification - Backpropagation Results



Tanh:

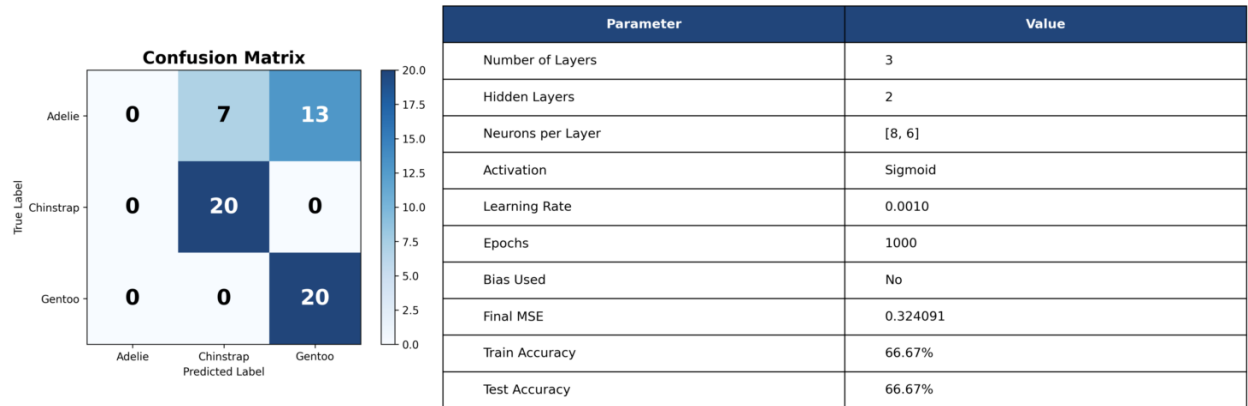
Penguin Species Classification - Backpropagation Results



- Trying 2 hidden Layers, Epochs = 1000, Learning Rate = 0.001 and without Bias.

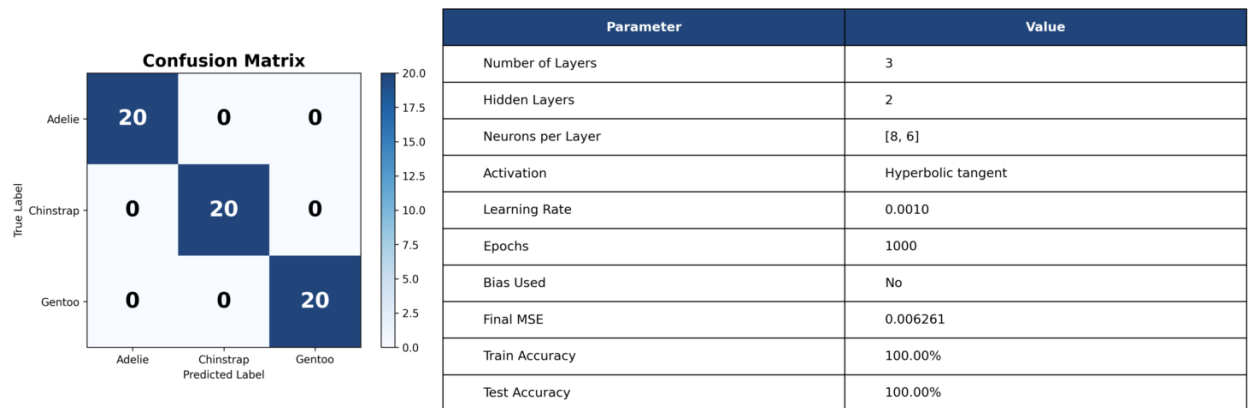
Sigmoid:

Penguin Species Classification - Backpropagation Results



Tanh:

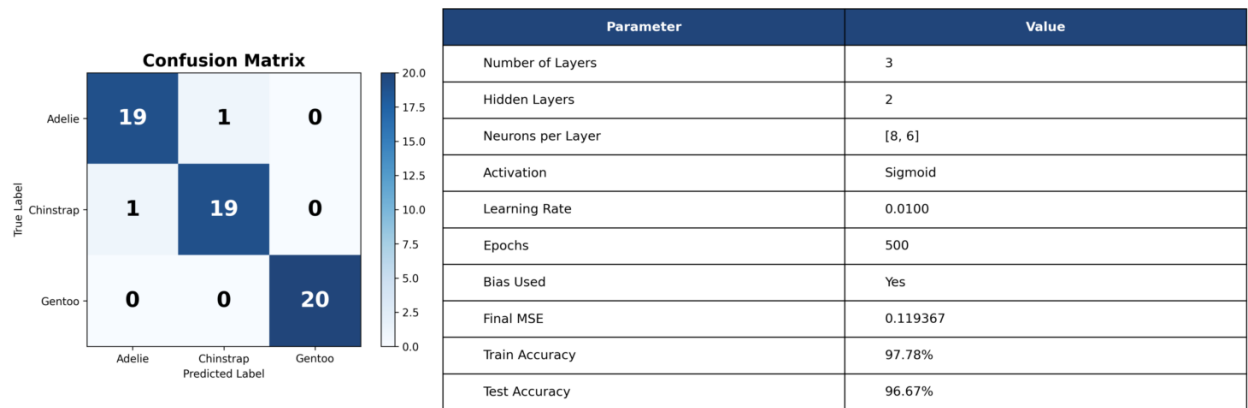
Penguin Species Classification - Backpropagation Results



- Trying 2 hidden Layers, Epochs = 500, Learning Rate = 0.01 and with Bias.

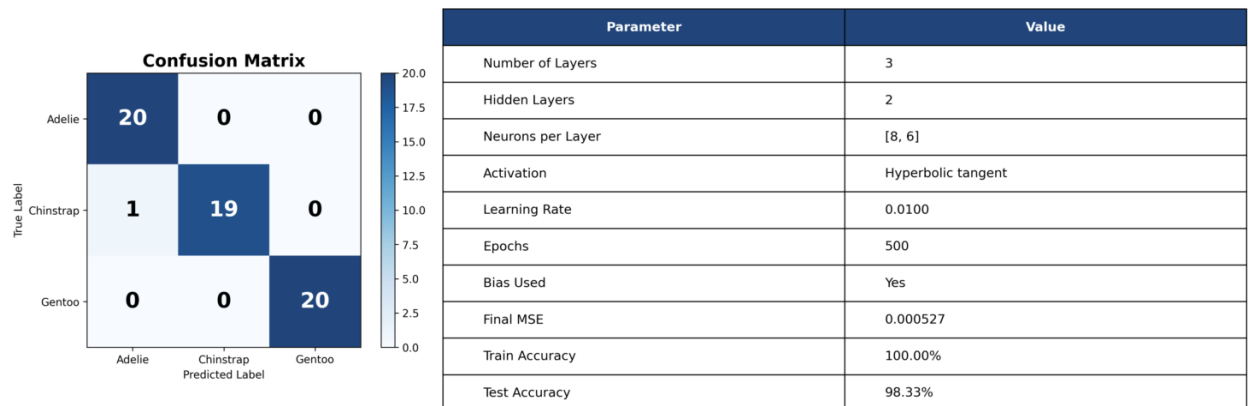
Sigmoid:

Penguin Species Classification - Backpropagation Results



Tanh:

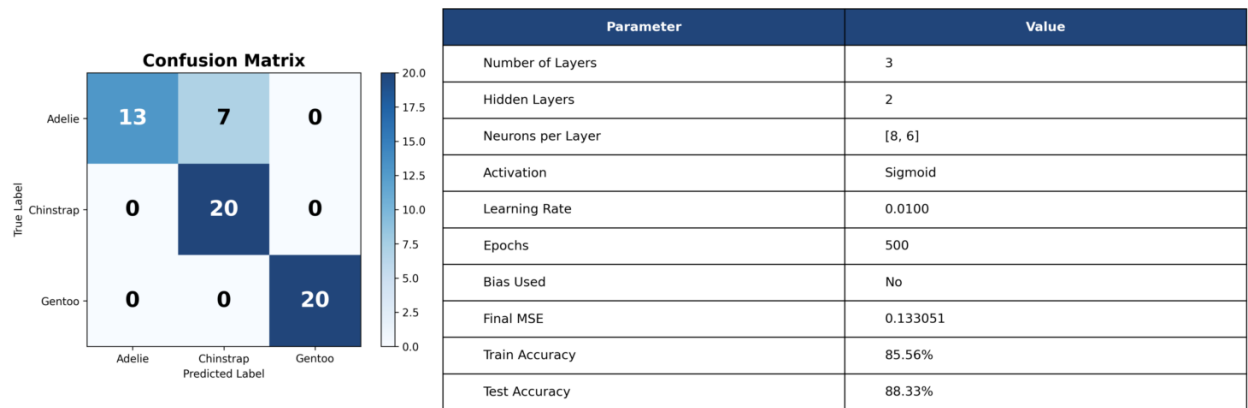
Penguin Species Classification - Backpropagation Results



- Trying 2 hidden Layers, Epochs = 500, Learning Rate = 0.01 and without Bias.

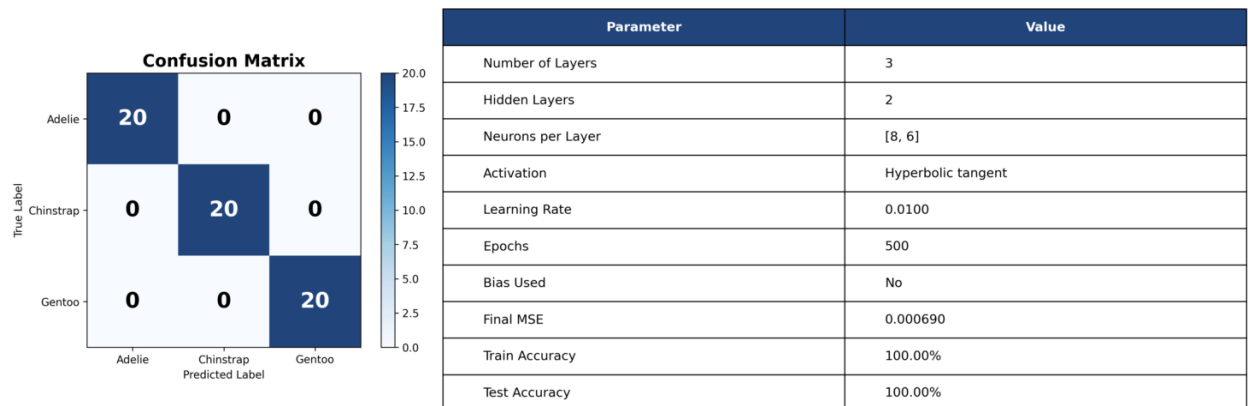
Sigmoid:

Penguin Species Classification - Backpropagation Results



Tanh:

Penguin Species Classification - Backpropagation Results



Conclusion:

- Adding Bias improves accuracy
- Using Tanh instead of Sigmoid activation function yielded better results